SIMATS ENGINEERING

CSA0705 - COMPUTER NETWORKS

STUDENT NAME : BALAJI A

REGISTER NO : 1911260245

DEPARTMENT : B.TECH COMPUTER SCIENCE ENGINEERING

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FACULTY NAME : KUMARAN G

NATURE OF ASSIGNMENT : INDIVIDUAL

1. Selection of Transmission Media

The choice depends mainly on **distance** and on the **noise** around the cable. Copper Ethernet (Cat6 / Cat6A) works only up to 100 m, so distance is the first thing to check.

Run	Distance	Chosen medium	Deciding reason
A. Robot cell near arc-welding machines	85 m	**Cat6A S/FTP** (shielded copper)	85 m is inside the 100 m copper limit, and the shielding (screened pairs plus overall screen) protects against the strong electrical noise of arc welding. Unshielded Cat6 UTP would pick up this noise.
B. Office floor to shop-floor switch	140 m	**Multimode fibre**	140 m is more than 100 m, so no copper cable (UTP or S/FTP) can be used. This is a short building run, so low-cost multimode fibre is enough, and fibre is not affected by electrical noise.
C. Shop floor to data centre	550 m	**Single-mode fibre**	Far beyond copper. Multimode fibre is limited at higher speeds (about 300–400 m at 10 Gbps), so single-mode fibre is the safe choice for 550 m. (Multimode is acceptable only if the link will stay at 1 Gbps.)
D. Plant to head office	400 km	**Leased line** (from a service provider)	400 km is a wide-area distance. A company cannot lay its own cable that far, so it rents a circuit from a carrier (which itself runs over long-distance fibre). Wireless is not reliable enough for this distance.

1.1 All six options compared for each run

Run	Cat6 UTP	Cat6A S/FTP	Multimode fibre	Single-mode fibre	Leased line	Wireless
A. 85 m, welding	No	**Best**	OK	OK	No	No
B. 140 m	No	No	**Best**	OK	No	Not preferred
C. 550 m	No	No	Limited	**Best**	No	Not preferred
D. 400 km	No	No	No	Only via a provider	**Best**	Not preferred

*Table 2: Suitability matrix. **Best** = chosen; **OK** = works but costs more or is more than needed; **Limited** = works only under conditions; **No** = not suitable.*

- **Run A:** fibre is also immune to welding noise, but it needs media converters and costs more. Shielded copper is enough for 85 m. Wireless is a poor choice near welding interference.
- **Run B:** extending copper with a switch in the middle would also work, but it adds a powered device on the shop floor. Fibre is cleaner.
- **Run D:** single-mode fibre is the physical medium inside the carrier's network, but the plant cannot use it without renting the service.

2. Copper Ethernet Distance Limit

A standard copper Ethernet channel (Cat6 or Cat6A) is limited to **100 m** (up to 90 m of fixed cable plus patch cords). Beyond this, the signal weakens (attenuation), crosstalk and noise grow, and the SNR becomes too low for reliable data.

Run	Distance	Within 100 m?	Comparison with the 100 m limit
A. Robot cell	85 m	Yes	15 m below the limit
B. Office to shop-floor switch	140 m	**No**	40 m over the limit (1.4 times)
C. Shop floor to data centre	550 m	**No**	450 m over the limit (5.5 times)
D. Plant to head office	400 km = 400,000 m	**No**	4,000 times the limit

Table 3: Distance check against the 100 m copper limit.

The run that clearly cannot use copper: D (400 km). Runs B (140 m) and C (550 m) also exceed 100 m.

- **Run D (400 km)** is the most impossible case: copper is not used for such wide-area links at all, so a leased line is needed.
- **Run B (140 m)** is the trap: it looks like an ordinary office cable, but it is 40 m longer than the copper limit, so copper cannot be chosen even though the run is short.
- **Run C (550 m)** is also far beyond the limit and needs fibre.
- Only **Run A (85 m)** is within 100 m, so it is the only run where copper is allowed.

3. Shannon Capacity Calculations (250 MHz Cable)

$$C = B \log_2(1 + SNR)$$
$$SNR\ (linear) = 10^{SNR(dB)\ /\ 10}$$

Given: bandwidth B = 250 MHz = 250×10^6 Hz.

3.1 Case 1: SNR = 40 dB

Step 1 – Convert dB to linear:

$SNR = 10^{40/10} = 10^4 = 10{,}000$

Step 2 – Find 1 + SNR:

$1 + 10{,}000 = 10{,}001$

Step 3 – Find the logarithm:

$\log_2(10{,}001) = \ln(10{,}001) \div \ln(2) = 9.2104 \div 0.6931 = 13.2879$

Step 4 – Find the capacity:

$C = 250 \times 10^6 \times 13.2879 = 3.322 \times 10^9$ bit/s

C (40 dB) ≈ 3,322 Mbps ≈ 3.32 Gbps

Step 1 – Convert dB to linear:

$SNR = 10^{15/10} = 10^{1.5} = 31.62$

Step 2 – Find 1 + SNR:

$1 + 31.62 = 32.62$

Step 3 – Find the logarithm:

$\log_2(32.62) = \ln(32.62) \div \ln(2) = 3.4850 \div 0.6931 = 5.0278$

Step 4 – Find the capacity:

$C = 250 \times 10^6 \times 5.0278 = 1.257 \times 10^9$ bit/s

C (15 dB) ≈ 1,257 Mbps ≈ 1.26 Gbps

3.2 Case 2: SNR = 15 dB (near welding machines

3.3 Comparison

Item	SNR = 40 dB	SNR = 15 dB (welding)
SNR (linear)	10,000	31.62
$\log_2(1 + \text{SNR})$	13.29	5.03
Capacity C	3.32 Gbps	1.26 Gbps
Capacity compared with 40 dB case	100 %	37.84 %

Table 4: Shannon capacity comparison.

Difference = 3,322 − 1,257 = 2,065 Mbps ≈ 2.07 Gbps
Ratio = 3,322 ÷ 1,257 = 2.64 times
Reduction = 100 − 37.84 = 62.16 %

What this means for the robot cell:

- Noise near the welding machines cuts the theoretical capacity to about 38 % of the quiet-cable value (a drop of about 62 %), even though the cable bandwidth is the same.
- Even at 15 dB, the theoretical capacity (1.26 Gbps) is still above 1 Gbps. So noise lowers the margin but does not remove the capacity.
- In practice, the noise also causes bit errors, which lead to resent frames and extra delay. This is why shielded cable (Cat6A S/FTP) is chosen for the robot cell.
- Capacity falls quickly as SNR falls, so protecting the cable from noise is more important than adding more bandwidth.

4. Why Actual Speed Is Below Shannon Capacity

Shannon capacity is an **upper limit**. It assumes ideal coding, ideal hardware and no extra bits. Real links always fall short because of the following reasons.

Factor	Why it lowers the actual speed
Protocol overhead	Headers, trailers (such as the FCS), preamble and gaps between frames carry no user data, so useful data (goodput) is less than the raw bit rate.
Noise	Shannon's value is calculated for a given SNR. Real noise varies (for example, bursts from welding), and extra margin must be kept, so systems run below the limit.
Encoding	Real line codes and modulation schemes cannot reach the ideal Shannon limit; some bandwidth is used for clock timing and signal shaping.
Error correction	Extra redundant bits are added so that errors can be detected or fixed. These bits use capacity but carry no new data.
Hardware limitations	Network cards, switches, connectors and cable quality are not perfect. Processing time and buffers also limit the speed.
Retransmissions	Corrupted or lost frames must be sent again, so the same data uses the link more than once.

Table 5: Reasons why real speed is below the Shannon capacity.

5. Telephone Line SNR

Find the SNR (in dB) needed for a telephone line with B = 3,100 Hz to carry C = 56 kbps = 56,000 bit/s. Rearrange Shannon's formula to find SNR:

$$C = B \log_2(1 + SNR)$$
$$C \div B = \log_2(1 + SNR)$$
$$SNR = 2^{C/B} - 1$$

$$SNR\ (dB) = 10 \log_{10}(SNR)$$

Step 1 – Find C ÷ B:

$C \div B = 56{,}000 \div 3{,}100 = 18.0645$

Step 2 – Find the linear SNR:

$SNR = 2^{18.0645} - 1 = 274{,}133 - 1 \approx 274{,}132$

Step 3 – Convert to dB:

$SNR\ (dB) = 10 \times \log_{10}(274{,}132) = 10 \times 5.4380 = 54.38$ dB

Required SNR ≈ 54.38 dB (about 274,000 times in power ratio)

Check: each doubling of SNR adds about 3.01 dB, so 18.0645 × 3.0103 = 54.38 dB, which agrees.

Is this realistic?

- **No, it is not realistic** for an ordinary analogue telephone line. An SNR of 54 dB is a very clean signal; voice-grade lines are commonly quoted at around 30–40 dB (general textbook value).
- For comparison, at 35 dB the same 3,100 Hz line has a Shannon limit of only about 3,100 × 11.63 ≈ 36 kbps.
- Real 56 kbps modems work only because one end of the connection is digital, which avoids some noise, and even then the speed is not guaranteed.

6. Alternative Concept Analysis – Wi-Fi 6

Given: B = 160 MHz = 160×10^6 Hz and SNR = 15 dB.

6.1 Shannon limit

Step 1: $SNR = 10^{15/10} = 31.62$

Step 2: $1 + SNR = 32.62$

Step 3: $\log_2(32.62) = 5.0278$

Step 4: $C = 160 \times 10^6 \times 5.0278 = 804.45 \times 10^6$ bit/s

Wi-Fi 6 Shannon limit ≈ 804.5 Mbps

This is about 64 % of the 250 MHz cable at the same 15 dB (804.45 ÷ 1,256.95), because the bandwidth is smaller (160 MHz instead of 250 MHz).

6.2 Is Wi-Fi 6 acceptable for robot commands with a 10 ms deadline?

Check 1: **Is capacity enough?** A small command frame is very short. Using the 70-byte frame from Task 3 (70 × 8 = 560 bits):

$$\text{Transmission time} = 560 \text{ bits} \div 804.45 \times 10^6 \text{ bit/s} \approx 0.70 \text{ µs}$$
$$\text{Deadline} = 10 \text{ ms} = 10{,}000 \text{ µs (about 14,000 times longer)}$$

So, on paper, the capacity is far more than enough. But the deadline cannot be judged by capacity alone.

6.3 Theoretical capacity vs real-time reliability

	Theoretical capacity	Real-time reliability
What it tells us	Maximum error-free bit rate for a given bandwidth and SNR	Whether every command arrives on time, every time
What it measures	Bits per second (an average, ideal limit)	Delay, jitter and loss in the worst case
Wi-Fi 6 value	≈ 804.5 Mbps	Not guaranteed by the Shannon limit
What affects it	Bandwidth and SNR only	Sharing the air with other devices, interference (including welding noise), retransmissions, contention, signal changes

Table 6: Capacity compared with real-time reliability.

Evaluation:

- Wi-Fi shares the air between many devices, so a command may wait for its turn. Interference or a weak signal causes lost frames, which are retransmitted and add delay.
- A 10 ms deadline needs the *worst-case* delay to stay below 10 ms every time. A high Shannon limit gives no such guarantee.
- The robot cell is near arc-welding machines, where the SNR is already low (15 dB) and noise changes unpredictably.
- **Conclusion:** Wi-Fi 6 is **not acceptable as the main link** for time-critical robot commands. It may be used for non-critical traffic such as monitoring, while the robot commands use the shielded Cat6A S/FTP cable chosen in Section 1.

7. Summary of Final Answers

Question	Final answer
Run A: robot cell, 85 m	**Cat6A S/FTP** – within 100 m, shielded against welding noise
Run B: office to switch, 140 m	**Multimode fibre** – over the 100 m copper limit
Run C: shop floor to data centre, 550 m	**Single-mode fibre** – long distance, high speed
Run D: plant to head office, 400 km	**Leased line** – wide-area distance
Runs that exceed 100 m copper limit	**B (140 m), C (550 m) and D (400 km)**; D cannot use copper at all
Shannon capacity, 250 MHz, 40 dB	≈ **3.32 Gbps**
Shannon capacity, 250 MHz, 15 dB	≈ **1.26 Gbps** (62.16 % lower)
SNR for 56 kbps on 3,100 Hz line	≈ **54.38 dB** – not realistic for a normal line
Wi-Fi 6 Shannon limit, 160 MHz, 15 dB	≈ **804.5 Mbps** – capacity is high, but real-time reliability is not guaranteed

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